



AFSS

BACKLOG TRACKER DOCUMENTATION

Australian Fire & Safety Systems — v1.0

Complete reference for the AFSS Backlog Tracker dashboard.

Dashboard · SimPRO Integration · Caching · Deployment · Automation

Next.js 16

React 19

TypeScript 5

Tailwind CSS v4

SimPRO API

Cloud Run

Red Adair Group · buoyant-purpose-475203-t9 · australia-southeast1

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The AFSS Backlog Tracker is an internal web dashboard that connects live to SimPRO to give management a real-time, consolidated view of the CFSP audit backlog across three companies, team availability, and scheduled technical support work.

KEY FEATURES

BACKLOG TABLE

- ▶ Live job counts, hours & invoice amounts per company
- ▶ Status rows: Scheduled / Tentative / Awaiting Client / Complete
- ▶ Month filter to view jobs by scheduled date
- ▶ Progressively loads — each company appears as data arrives

TECHNICAL TEAM SUPPLY

- ▶ Remaining billable hours per auditor for current month
- ▶ Leave automatically deducted from SimPRO schedule data
- ▶ Before 3 PM: today counts; after 3 PM: tomorrow starts

TECHNICAL SUPPORT WORKS

- ▶ Other billable work scheduled to tech team (today → month end)
- ▶ Invested time: pending jobs for customers being served this period
- ▶ Quality Assurance jobs assigned to QA Officer

PERFORMANCE & AUTOMATION

- ▶ 1-hour disk cache — fast responses after warmup
- ▶ Cloud Scheduler warms cache every 30 minutes automatically
- ▶ Stale-while-revalidate — always serve instantly, refresh in background
- ▶ "Refresh now" button triggers immediate background refresh

TECHNOLOGY STACK

Category	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript 5
Styling	Tailwind CSS v4
UI Library	React 19
Data Source	SimPRO REST API v1.0
Cache Storage	/tmp JSON files (OS temp dir)
Hosting	Google Cloud Run
Region	australia-southeast1 (Sydney)
Build	Google Cloud Build + Docker
Scheduler	Google Cloud Scheduler

LIVE SITE

Production URL:

`https://bryan-technical-afss-712513641417.australia-southeast1.run.app`

GCP PROJECT

Project ID: `buoyant-purpose-475203-t9`

Service name: `bryan-technical-afss`

Region: `australia-southeast1`

The main dashboard table shows job counts, estimated hours, and invoice amounts broken down by status row and company column. All numbers update automatically from SimPRO.

Status Rows

Colour	Row Name	Condition	Notes
■	Scheduled Awaiting to be Done	Job has a confirmed scheduled date in SimPRO	Date sourced from schedule blocks or job detail fields
■	Awaiting Client Info	Status Name contains "client" or "awaiting"	Pending jobs stuck waiting on customer action
■	Tentative Awaiting Scheduling	Pending job with no confirmed schedule date	Default bucket for unscheduled CFSP audit jobs
■	Attendance Complete / Results To Be Released	Stage = Progress (non-AE Evac), or status name contains complete/released/attendance	AE Evac Progress-stage jobs treated as Scheduled, not Complete

Month Filter

The dropdown (top-right) filters all status rows to only show jobs whose **scheduled date** falls in the selected month. Selecting a month also includes the **previous month** in results (e.g. selecting June shows May + June combined). Defaults to "All".

TOTAL BACKLOG ROW

The "Total Backlog as at end of period" row sums every job across all status rows and all three companies — regardless of the month filter.

AE EVAC SPECIAL RULE

AE Evac (company 10) uses the **Progress stage** to mean "booked" rather than "attendance done." Progress jobs for AE Evac are placed in **Scheduled** if they have a date, or **Tentative** if they do not — not in the Complete row like all other companies.

Loading & Partial Data States

The dashboard uses a **progressive loading** strategy — each of the 6 company/stage combinations renders as soon as its data arrives, rather than waiting for all 6.

LOADING INDICATORS

- ▶ **Loading...** (blue, top bar) — fetches are still in flight
- ▶ **Refreshing schedule data...** (amber) — data arrived but schedule enrichment is still running in background
- ▶ — **dashes** in Scheduled & Tentative rows — enrichment not yet complete; correct split unknown until schedule data is ready
- ▶ **Updated: HH:MM:SS** — timestamp of the last successful data fetch

NORMAL OPERATION

With the cache warm (Cloud Scheduler runs every 30 min), the dashboard loads with correct numbers in **under 2 seconds**. The dashes only appear on a cold cache (new deploy or instance restart).

Technical Team Supply

Shows the remaining billable supply hours for each of the three AFSS auditors for the **current calendar month**.

Column	Description
May (current month)	Remaining working days × 8 hrs
Total Supply Hours	Same — matches the SimPRO "end of period" concept
Roles	Each auditor's primary function description

WORKING DAYS CALCULATION

REMAINING DAYS LOGIC

- ▶ Before 3:00 PM today → today counts as a remaining day
- ▶ From 3:00 PM onwards → today is excluded (day is considered done)
- ▶ Weekends (Sat/Sun) are never counted
- ▶ Counts from start date through the last day of the current month

LEAVE DEDUCTION

Leave is sourced from the SimPRO schedule API. Activity blocks with reference ID **1** (Annual Leave) or **2** (Sick/Personal Leave) are detected and the note at the bottom of the table lists any leave for today or upcoming in the current month.

LEAVE DATA SOURCE

The `/api/leave` endpoint scans all weekdays of the current month plus 7 days ahead (to catch pre-booked upcoming leave), fetching schedule blocks in parallel for speed.

Technical Support Works

Three columns of automated statistics, each sourced live from SimPRO with a 1-hour cache.

OTHER BILLABLE WORK SCHEDULED TO TECH TEAMS

- ▶ Rate: \$100 per hour
- ▶ Date range: **Today** → **end of current month** (rolling daily)
- ▶ Scans the SimPRO schedule day-by-day for job-type blocks
- ▶ Staff: Josh Roger, Muhammad Soban, Ryan Gordon + Tentative placeholders
- ▶ **Excludes** CFSP audit jobs (tech ID 1126) and "system testing" tagged jobs

INVESTED TIME (TRAINING / NON-BILLABLE)

- ▶ Rate: \$100 per hour
- ▶ Date range: **Today** → **end of current month**
- ▶ Finds customers with jobs scheduled to the team in this period
- ▶ Counts all pending SimPRO jobs belonging to those customers
- ▶ Represents pending workload of clients the team is actively serving

QUALITY ASSURANCE — OVERALL TOTAL IN SIMPRO

- ▶ Rate: \$100 per hour
- ▶ **No date filter** — shows full current QA workload
- ▶ All Pending + Progress jobs in company 1
- ▶ Filtered to jobs assigned to "A Quality Assurance Officer" technician

CFSP Filter & Job Classification

A job qualifies as a CFSP audit job if technician ID `1126` ("A CFSP ONLY") is assigned to it. This filter applies to companies 1 and 10. Company 8 (CHUBB/AFAC) returns all jobs without filtering.

SCHEDULED DATE DETECTION

A job is classified as "Scheduled" if any of these data points exist:

Priority	Source	Details
1st	Schedule blocks API	Per-job schedule blocks — most accurate, includes actual booking date
2nd	Job detail fields	Checks <code>Scheduled</code> , <code>DateScheduled</code> , <code>ScheduledDate</code> , <code>DateBooked</code>

CALCULATION DEFAULTS

Field	Primary Source	Default if Missing
Estimated Hours	<code>Totals.ResourcesCost.LaborHours.Estimate</code>	2 H
Invoice Amount	<code>Total.ExTax</code>	\$3
AE Evac rate	Header label only	\$3

Job Enrichment Pipeline

Raw job list data from SimPRO only contains basic fields. The enrichment pipeline adds site details, confirmed schedule dates, and customer group classification per job.

- 1 Fetch job list (pages 1–10 in parallel)
Fetches up to 2,500 jobs per company/stage. Filters to CFSP tech ID 1126 (except company 8).
- 2 Parallel enrichment — 4 streams at concurrency 10
Details: `/jobs/{id}` — labour hours, status, all fields
Sites: `/sites/{id}` — site name and address
Schedules: `/schedules/?JobID={id}` — confirmed date + hours
Customers: `/customers/{segment}/{id}` — customer group
- 3 Merge & write cache
Enriched jobs are merged with `_site`, `_scheduledDate`, `_scheduledHours`, and `_customerGroup` fields, then written to `/tmp` as JSON.

ENRICHMENT TIME

Enrichment takes **30–160 seconds** per company/stage depending on job count and SimPRO API speed. For 85 jobs (company 1 Pending), expect ~45 seconds. This only occurs on a cold cache — normal operation serves cached data instantly.

RATE LIMITING

SimPRO enforces API rate limits (HTTP 429). All API calls use **exponential backoff** with up to 8 retry attempts. Initial retry delay is 1 second, doubling each attempt (1s → 2s → 4s → 8s...). Concurrent requests are capped at **10 per pool** to reduce 429 frequency.

All Endpoints

Method	Endpoint	Cache TTL	Description
GET	/api/data?company=N&stage=S	1 hour	Job list for one company/stage. Returns enriched JSON. Serves cache if fresh.
GET	/api/data?company=N&stage=S&force=1	—	Force refresh. Returns partial data immediately with <code>X-Partial: 1</code> , triggers background re-enrichment.
GET	/api/leave	1 hour	Leave schedule for all 3 team members for the current month. Returns grouped consecutive leave ranges.
GET	/api/leave?force=1	—	Force refresh leave cache. Scans SimPRO schedules for current month + 7 days ahead.
GET	/api/tech-support	1 hour	Technical Support Works stats (other billable, invested time, QA). Returns 3 stat objects.
GET	/api/tech-support?force=1	—	Force refresh tech-support cache. Returns stale immediately, refreshes in background.
GET	/api/warmup	—	Pre-warms all caches: 6 job combos sequentially + leave + tech-support in parallel. Max 300s.

Enriched Job Response Fields

Field	Source	Description
<code>_site</code>	Sites API	Full site object (name, address)
<code>_scheduledDate</code>	Schedules API	ISO date of first confirmed schedule block
<code>_scheduledHours</code>	Schedules API	Total hours across all schedule blocks
<code>_customerGroup</code>	Customers API	Customer group name (e.g. "Government")
<code>_company</code>	Dashboard	Company ID added client-side (1, 8, or 10)

X-PARTIAL: 1 HEADER

When the response contains jobs **not yet enriched** with schedule, site, and customer data, the API returns this header. The dashboard detects it and automatically retries every **5 seconds** until the header is absent and full data is available.

Leave API Response Shape

```
// Array — one entry per team member [ { "id": 1581,
"name": "Muhammad Soban", "role": "Primary APFS",
"monthlyHours": 32, "leave": [ { "from": "2026-05-04", "to": "2026-05-18" } ] }, // ... Ryan Gordon,
Josh Roger ]
```

Tech Support API Response Shape

```
{ "otherBillable": { "jobs": 10, "hours": 17.75,
"amount": 1775 }, "investedTime": { "jobs": 37,
"hours": 70.58, "amount": 7058 },
"qualityAssurance": { "jobs": 4, "hours": 8.0,
"amount": 800 } }
```

All three data sources (job data, leave, tech-support) use an identical caching pattern: write enriched JSON to `/tmp` on first fetch, then serve from cache for 1 hour while refreshing in the background when stale.

Cache Storage & TTL

Cache	File	TTL
Job data (per combo)	<code>afss-jobs-{co}-{stage}.json</code>	1 hour
Leave schedule	<code>afss-leave-cache.json</code>	1 hour
Tech Support stats	<code>afss-tech-support-v18-cache.json</code>	1 hour

CLOUD RUN EPHEMERAL DISK

The `/tmp` disk is **wiped when Cloud Run recycles the container** (e.g. on new deployment, idle scale-down, or instance restart). After a wipe, the next user request triggers a cold cache path — the Cloud Scheduler job ensures the cache is re-warmed within 30 minutes.

Stale-While-Revalidate Flow

- 1

Request arrives

Check `/tmp` for existing cache file.
- 2

Cache is fresh (age < 1 hour, not partial)

Return cached data immediately. No SimPRO call needed.
- 3

Cache is stale or partial

Return stale/partial data immediately (fast), then trigger background re-enrichment from SimPRO. Next request gets fresh data.
- 4

No cache (cold start)

Fetch job list immediately (fast path), return with `X-Partial: 1`. Background enrichment starts. Dashboard retries every 5 s.

Cold Cache — What the User Sees

Time	Dashboard State
0–2s	Fast-path data arrives — job count and totals visible. Scheduled & Tentative rows show —
2–5s	All 6 company/stage responses in. "Refreshing schedule data..." amber indicator shows.
5–120s	Dashboard retries every 5 s. Background enrichment is running.
30–160s	Enrichment completes. Dashboard gets full data — correct Scheduled / Tentative split appears.

Warmup Endpoint Performance

Company / Stage	Typical Time
Company 1 — Pending (85 jobs)	~45–160 s
Company 1 — Progress	~5 s
Company 8 — Pending	~15–65 s
Company 8 — Progress	< 1 s
Company 10 — Pending	< 1 s
Company 10 — Progress (123 jobs)	~52–96 s
Total (sequential)	~2–5 min

WHY SEQUENTIAL?

Company/stage combos are warmed one-at-a-time to avoid hammering SimPRO with 60+ concurrent API calls (which causes heavy 429 rate-limiting and slows everything down). Leave and tech-support caches are warmed in parallel alongside the sequential job warmup.

AFSS Auditor Team

Name	SimPRO Staff ID	Role	Leave Tracking
Muhammad Soban	1581	Primary APFS	Annual Leave (ref 1) + Sick/Personal (ref 2)
Ryan Gordon	15	Mixed of Technical Support Works and Secondary APFS	Annual Leave (ref 1) + Sick/Personal (ref 2)
Josh Roger	1753	Estimation / Office Management Time ETC	Annual Leave (ref 1) + Sick/Personal (ref 2)

Companies & Job Classification

Dashboard Column	Company ID	SimPRO Company	Filter Applied	Default Rate
RM AFSS	1	Company 1	Jobs with tech ID 1126 (A CFSP ONLY)	\$330 / job or actual ExTax
CHUBB/AFAC AFSS	8	Company 8	All jobs (no technician filter)	\$330 / job or actual ExTax
AE Evac Procedure Audits	10	Company 10	Jobs with tech ID 1126 (A CFSP ONLY)	\$385 / hour (shown in header)

Stages Fetched per Company

Each company fetches two job stages from SimPRO, giving **6 total combinations** that are cached independently.

Stage	Dashboard meaning
Pending	Jobs yet to be attended — Scheduled, Tentative, or Awaiting Client Info
Progress	Jobs attended / in progress — Attendance Complete for all except AE Evac

AE EVAC EXCEPTION

AE Evac uses **Progress** to mean "booking confirmed" rather than "attendance done." The dashboard routes AE Evac Progress jobs to Scheduled (if date exists) or Tentative (if no date), not to the Complete row.

SimPRO API Connection

All data is fetched from the SimPRO REST API v1.0 using a Bearer token stored as a Cloud Run secret environment variable.

API USAGE PATTERN

- Job list: `GET /api/v1.0/companies/{id}/jobs/?pageSize=250&columns=...`
- Job detail: `GET /api/v1.0/companies/{id}/jobs/{jobId}`
- Schedules: `GET /api/v1.0/companies/{id}/schedules/?JobID={id}`
- Sites: `GET /api/v1.0/companies/{id}/sites/{siteId}`
- Customers: `GET /api/v1.0/companies/{id}/customers/{segment}/{id}`
- Staff: `GET /api/v1.0/companies/1/staff/?pageSize=250&columns=ID,Name`
- Day schedules: `GET /api/v1.0/companies/1/schedules/?Date=YYYY-MM-DD`

TOKEN NOTE

The `SIMPRO_TOKEN` env var is stripped of any leading BOM characters () and whitespace on startup to avoid authentication failures from copy-paste artifacts.

Cloud Run Setup

Setting	Value
Service name	bryan-technical-afss
Region	australia-southeast1
GCP Project	buoyant-purpose-475203-t9
Project Number	712513641417
Max duration	300 s (warmup endpoint)
Build method	Dockerfile via Cloud Build

Deploy Command (PowerShell)

```
gcloud run deploy bryan-technical-afss --source . \
  --region australia-southeast1 --project buoyant-
  purpose-475203-t9 --quiet
```

BUILD TIME

Cloud Build typically takes **4–8 minutes** to build the Docker image and deploy the new revision. The old revision stays live until the new one passes health checks.

Post-Deploy Cache Warmup

Deploying a new revision creates a fresh container — the `/tmp` cache is empty. Run the warmup command immediately after deploy:

```
# PowerShell — takes 2-5 minutes Invoke-WebRequest `
-Uri "https://bryan-technical-afss-712513641417
.australia-southeast1.run.app/api/warmup" -
TimeoutSec 310
```

Post-Deploy Checklist

- 1 Deploy completes (gcloud run deploy)**
Confirm the new revision URL appears and "Routing traffic...done" is shown.
- 2 Run warmup endpoint**
Use the `Invoke-WebRequest` command with `-TimeoutSec 310`. Verify all 6 results show `"ok": true`.
- 3 Open the dashboard**
Navigate to `/dashboard` — should load with correct data in under 2 seconds, no amber "Refreshing" indicator.
- 4 Verify Scheduled row shows real numbers**
If the Scheduled and Tentative rows show actual numbers (not dashes), enrichment is complete and the cache is warm.

TypeScript Check Before Deploying

Always run the type checker before deploying to catch compile errors early:

```
# In the project directory npx tsc --noEmit # No
output = no errors. Safe to deploy.
```

LOCAL DEVELOPMENT

Run `npm run dev` to start the local server on `http://localhost:3000`. Set `SIMPRO_BASE_URL` and `SIMPRO_TOKEN` in a `.env.local` file. The local dev cache also writes to `/tmp`.

All 6 company/stage combos, leave, and tech-support will be cached and ready before the first user visits the dashboard.

Cloud Scheduler — Automatic Cache Warming

A Cloud Scheduler job (`afss-warmup-job`) automatically hits the warmup endpoint every 30 minutes, ensuring the cache is always fresh before the 1-hour TTL expires.

Setting	Value
Job name	<code>afss-warmup-job</code>
Schedule	<code>*/*30 * * * *</code> (every 30 min, UTC)
Region	<code>australia-southeast1</code>
Target URL	<code>/api/warmup</code>
HTTP method	GET
Attempt deadline	310 seconds
State	ENABLED

WHY 30 MINUTES?

The cache TTL is 1 hour. Warming every 30 minutes gives a comfortable buffer — even if one warmup run is slow or fails, the cache is still valid from the previous successful run. Users almost never see the slow enrichment path.

Environment Variables

Variable	Description
<code>SIMPRO_BASE_URL</code>	SimPRO instance base URL — e.g. <code>https://example.simpro.co</code>
<code>SIMPRO_TOKEN</code>	SimPRO API Bearer token. Leading BOM characters are stripped automatically on startup.

SECURITY

Both variables are stored as **Cloud Run environment variables** — never committed to the repository or included in the Docker image. To update them, use the GCP Console or `gcloud run services update` with `--set-env-vars`.

Project File Structure

```
afss-backlog/ |— app/ # Next.js App Router | |—  
api/ | |— data/ # Job data endpoint | | |—  
route.ts | |— leave/ # Leave schedule endpoint |  
| | |— route.ts | |— tech-support/ # Tech  
Support Works endpoint | | |— route.ts | |—  
warmup/ # Cache pre-warm endpoint | |— route.ts |  
|— dashboard/ # Dashboard page | |— page.tsx #  
Main dashboard UI | |— lib/ | |— simpro.ts #  
SimPRO API client + cache | |— layout.tsx # Root  
layout |— page.tsx # Backlog list page |—  
Dockerfile # Cloud Run container build |—  
next.config.ts # Next.js config |— tsconfig.json #  
TypeScript config |— package.json
```

KEY SOURCE FILES

File	Purpose
<code>app/lib/simpro.ts</code>	SimPRO API client, pooled fetching, disk cache read/write, <code>warmAll()</code>
<code>app/api/data/route.ts</code>	Job data API — smart cache, fast path, X-Partial header
<code>app/api/leave/route.ts</code>	Leave detection — scans SimPRO schedules, groups consecutive dates
<code>app/api/tech-support/route.ts</code>	Tech Support Works — 3 independent stats from SimPRO schedule & jobs APIs
<code>app/api/warmup/route.ts</code>	Orchestrates all cache warming — called by Cloud Scheduler every 30 min
<code>app/dashboard/page.tsx</code>	Full dashboard UI — progressive loading, partial detection, auto-retry